

# The $\cos 2\phi$ modulation on tensor-polarized ${}^6\text{Li}$ : projected inclusive and coherent measurements

2026-08-17 · development run 5 · polligen (evgen/) + polli\_fastsim (fastsim/) · companions: plans/06 (background budget), docs/note\_cos2phi\_coherent\_6Li.md, reports/polarized\_li\_primer.html (pedagogical introduction) · external numbers verified against primary sources; this revision adds dedicated fact checks of the moment constraint and of the far-forward detection chain

This note presents two projected EIC measurements of the  $\cos 2\phi$  azimuthal modulation on a transversely tensor-polarized  ${}^6\text{Li}$  beam: inclusive DIS, requiring only the scattered electron, and coherent diffraction with the  ${}^6\text{Li}$  detected intact in its  $1^+$  ground state. The modulation arises from photon double helicity flip and probes gluon transversity, to which neither quarks nor bound nucleons contribute at leading twist [1,2]; it has not been measured on any target. The projections are given together with the assumptions entering each number and the checks performed on them.

**13.5%**

coherent-tag acceptance at IP6 with the 0.20 GeV envelope cut,  $\exp(-B \cdot p_T^2 / \text{cut})$ ,  $B = 50 \text{ GeV}^{-2}$  [9–20% for  $B \in 40\text{--}60$ ]. The cut is documented for top-rigidity beams; for  ${}^6\text{Li}$  it is an optics assumption (§4.1).

**$1.1 \times 10^7 / 10^8$**

tagged coherent events in 1 / 10 EIC years (scenario  $f_0 = 0.04$ ); best-bin  $\delta A = 1.9 \times 10^{-3} / 6 \times 10^{-4}$ , i.e.  $5\sigma$  floors at 1.0% / 0.3% modulation

**0.036**

deformation-anchored ( $a_2$ ) of the tagged sample at  $P_{zz} = 0.6$  [band 0.018–0.059; one-sided  $\times 0.73$  rate-weighting systematic], scaled from the polarized-deuteron CGC calculation [13]; sign flip vs the deuteron predicted

## 1 • The observable

For unpolarized electrons on a spin-1 target whose spin axis makes angle  $\theta_m$  with the beam, the inclusive cross section per spin projection  $m$  is [1,2]

$$\frac{d\sigma^{(m)}}{dx dy d\phi} = \frac{2y\alpha^2}{Q^2} \left[ F_1 + \frac{2}{3} a_m b_1 + \frac{1-y}{xy^2} \left( F_2 + \frac{2}{3} a_m b_2 \right) - \frac{1-y}{y^2} c_m \sin^2 \theta_m \Delta(x, Q^2) \cos 2\phi \right]$$

$$a_m = \frac{1}{4} c_m (3 \cos^2 \theta_m - 1), \quad c_m = 3|\lambda_m| - 2 \rightarrow (1, -2, 1)$$

The  $\cos 2\phi$  term requires the virtual photon to flip its helicity by two units. Since no spin- $1/2$  constituent can absorb two units of helicity, neither quarks nor bound nucleons contribute at leading twist:  $\Delta(x, Q^2)$  is purely gluonic and vanishes for a nucleus composed of nucleons alone [1]. At leading order it is generated by the gluon transversity distribution  $\delta G$  through the  $O(\alpha_s)$  photon–gluon convolution

$$\Delta(x, Q^2) = -\frac{\alpha_s(Q^2)}{2\pi} \left( \sum_q e_q^2 \right) x^2 \int_x^1 \frac{dy}{y^3} \delta G(y, Q^2)$$

(in the sign convention of Ref. [25]; conventions differ between papers).  $\Delta$  has not been measured for any target; the only prior experimental document is a JLab letter of intent without a sensitivity projection [3]. Quantitative expectations are scarce. The only moment estimate is the Sather–Schmidt bag-model calculation for the spin- $3/2$   $\Delta^{++}$  baryon,  $\int x \Delta dx = -0.012 \alpha_s$  [4], which suggests asymmetries at the per-mille level; percent-level values arise only in maximal-transversity ( $\Delta_T g = \Delta g$ ) scenarios for proton–deuteron Drell–Yan, described by their authors as optimistic [5]. The projections therefore span the scenario band  $\Delta/F_1 \in 10^{-3}\text{--}10^{-2}$ .

The measured amplitude and its inversion back to the structure function are

$$A^{\cos 2\phi} = -\frac{1-y}{y^2} \frac{\Delta}{D_\phi}, \quad D_\phi = F_1 + \frac{1-y}{xy^2} F_2, \quad \hat{\Delta} = \hat{A} \frac{y^2 D_\phi}{1-y}$$

The fastsim approximation set is  $A_{//} = D \cdot g_1 / F_1$  (the  $y^2$ -suppressed term dropped),  $g_2 = g_2^{\text{WW}}$ , and no target-mass corrections. The last matters only at high  $x$  and low  $Q^2$ ; the sweet-spot bins sit at  $x \leq 0.06$ , where  $y^2 = 4M^2 x^2 / Q^2 < 10^{-2}$ , so it is a labeling caveat rather than a bias. Structure functions are toy parameterizations with

PDF-grid (CT18NLO/NNPDFpol11) backends behind the same interface: rates are reliable to a factor of  $\approx 1.5$ , and asymmetry shapes are scenarios by construction.

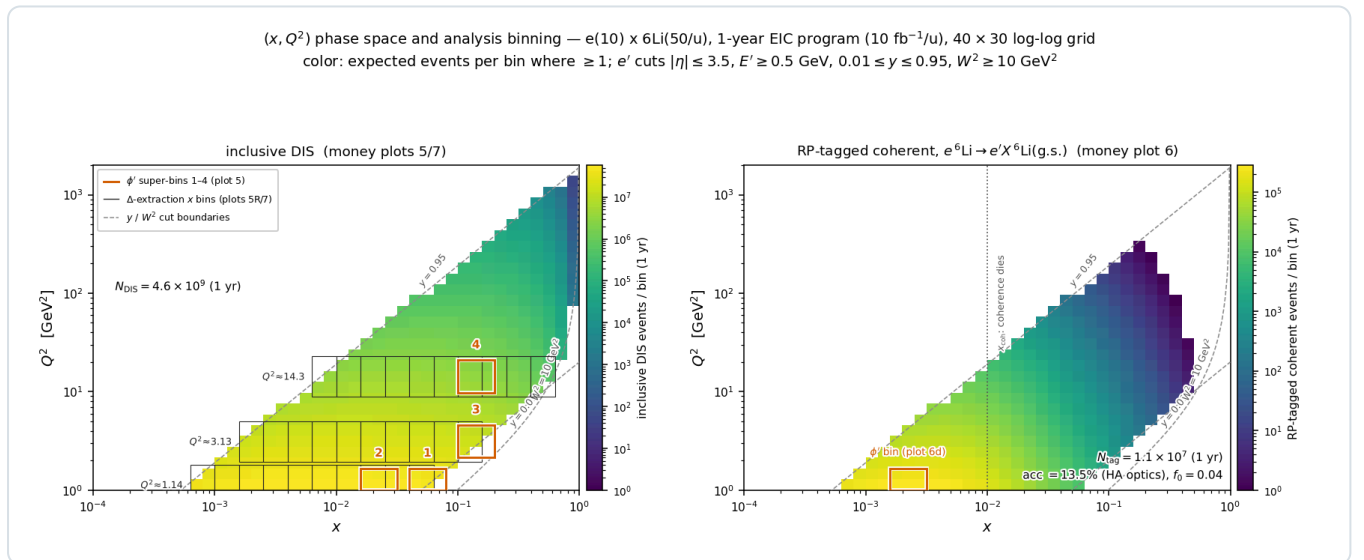
## 2 • Machinery, statistics, and binning

**Generator.** No public generator produces polarized e+A events for  $A > 1$  (BeAGLE is unpolarized; DJANGO and PEPSI are nucleon-level and longitudinal-only). polligen therefore implements only the missing component, the spin-density matrix  $\otimes$  structure-function master formula with spin-correlated sampling, and relies on existing tools for the rest. Its validation gates are bin-wise identity with the analytic asymmetry formulas at relative tolerance  $10^{-12}$ , pseudo-experiment spreads that close on every analytic error map, and unbiased  $\cos 2\phi$  recovery under  $\phi$ -acceptance holes.

**Full-luminosity statistics.** For a binned estimator, drawing each  $\phi$ -bin count from a Poisson distribution with the exact expected yield is statistically identical to event-level sampling, so a full-luminosity pseudo-experiment costs 24 Poisson draws ( `sample.phi_histogram_pseudo` ). Event-level sampling remains available and is what the closure tests use.

**Estimator.** The amplitude comes from a binned least-squares fit of  $N(\phi') = C(1 + A \cos 2\phi')$ , not the  $2\langle \cos 2\phi' \rangle$  moment: with realistic  $\phi$  holes the moment is biased by more than the signal, while the fit stays unbiased. Errors follow  $\delta A = \sqrt{(2/N)/P_{ZZ}}$ , validated by pseudo-experiments. The fit carries no  $\cos \phi'$  term; the m-symmetric tensor fills set  $a_1 = 0$  by construction.

**Binning.** The four displayed (x,  $Q^2$ ) super-bins are selected by statistical significance within  $Q^2$  bands: an unconstrained ranking concentrates at the lowest  $Q^2$ , while the  $Q^2$  lever arm provides the consistency check against power-suppressed contributions (§6). Each super-bin merges  $3 \times 3$  cells of the  $40 \times 30$  analysis grid. Rates use per-nucleon luminosity,  $F_2^A/A$ , scattered-electron cuts  $|\eta| \leq 3.5$ ,  $E' \geq 0.5$  GeV,  $0.01 \leq y \leq 0.95$ ,  $W^2 \geq 10$  GeV<sup>2</sup>, and the full luminosity in the transverse-tensor fill; projections are quoted for the 1-year ( $10 \text{ fb}^{-1}/u$ ) and 10-year ( $100 \text{ fb}^{-1}/u$ ) programs.



**Phase-space coverage and analysis binning** ( $e^{10} \times {}^6\text{Li}50$ , 1-year program,  $40 \times 30$  log-log grid; color: expected events per bin where  $\geq 1$ ; dashed lines:  $y$  and  $W^2$  cut boundaries). **Left, inclusive DIS** ( $4.6 \times 10^9$  accepted events in year 1): boxes 1–4 are the  $\phi'$  super-bins of money plot 5; gray combs are the three  $Q^2$  slices ( $Q^2 \times [1/1.6, 1.6]$ ) whose merged x-bin pairs form the data points of the amplitude-versus- $x$  panel and of the  $\Delta$  extraction (plots 5R/7), retained where  $N \geq 10^3$  and  $\delta A \leq 8 \times 10^{-3}$ . **Right, RP-tagged coherent channel:** the same map multiplied by  $f_{\text{coh}}(x)$  and the 13.5% tag acceptance.  $N_{\text{tag}} = 1.1 \times 10^7$  in year 1; 98% of the tagged sample lies at  $x < 0.03$  (median  $3.5 \times 10^{-3}$ ). The box is the super-bin of money plot 6(d); plot 6(c) is this map summed over  $Q^2$ .

## 3 • The $\Delta$ model and the moment constraint

**Provenance.** Sather and Schmidt computed the lowest moment of  $\Delta$  in the MIT bag model for the spin-3/2  $\Delta^{++}$  baryon, together with its  $Q^2$  evolution; they provide no  $x$  dependence [4]. The explicit  $\alpha_s(Q^2)$  is the leading-

order Wilson coefficient of the photon–gluon box, so the  $O(\alpha_s)$  suppression relative to  $F_1$  is model-independent; the value  $-0.012$  is the bag estimate. Lattice QCD provides two related results, each with substantial caveats: the bare  $\phi$ -meson matrix element  $A_2 = 0.23(2)(5)$  at  $m_\pi = 450$  MeV [25] converts to  $\int x \Delta|_\phi \approx -0.006 \alpha_s$  (our conversion, unrenormalized), while the NPLQCD deuteron calculation gives a  $\delta G$  moment of  $-0.010(3)$ , bare at  $m_\pi \approx 806$  MeV [27], a transversity-to-momentum ratio roughly ten times below that of the  $\phi$  and implying the opposite sign for the deuteron's  $\Delta$  moment. No calculation of  $\Delta$  exists for  ${}^6\text{Li}$  or any  $A > 2$  nucleus; the only nuclear model is a  $\Delta\Delta$ -isobar estimate for the deuteron [34]. The literature thus brackets the nuclear  $\Delta$  in both directions: suppressed, as in the NPLQCD deuteron, or enhanced by binding in analogy with the EMC effect, as argued in the polarized-ion white paper [50].

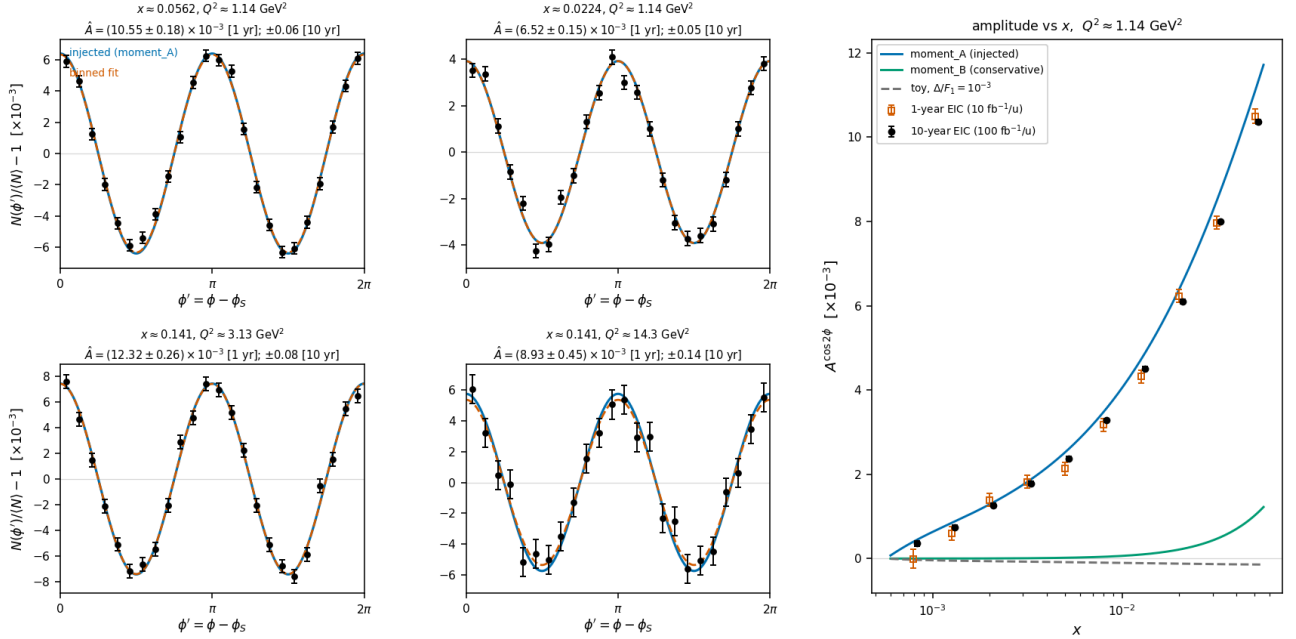
**Implementation.** All  $\Delta$  models reside in one registry ( `polli_fastsim/delta_models.py` ). The production default, `moment_A` , is an ansatz normalized to the bag moment applied to the per-nucleon  $\Delta$  of  ${}^6\text{Li}$ :

$$\Delta_A(x, Q^2) = \frac{1}{3} A \alpha_s(Q^2) F_1(x, Q^2) \frac{x^a(1-x)^b}{N_{\text{peak}}}, \quad \int_0^1 x \Delta(x, \langle Q^2 \rangle) dx = -0.012 \alpha_s \Rightarrow A = -0.294$$

with  $(a, b) = (0.7, 3)$ ; the factor  $1/3$  restricts the exotic glue to the polarized deuteron-like pair, while the whole-nucleus  $P_{ZZ}$  remains a separate factor, so that no double counting occurs. The constraint is imposed at the rate-weighted  $\langle Q^2 \rangle = 3.9$  GeV<sup>2</sup> of the accepted phase space; because  $F_1$  evolves, the realized moment varies between  $-4\%$  and  $+7\%$  across the plotted  $Q^2$  slices.  $\alpha_s$  is taken from the CT18NLO table (the leading-order analytic fallback is 14% higher). The solved  $A = -0.294$  is consistent with the EPPS21-based `money_delta` value of  $-0.31$  at its  $\langle Q^2 \rangle = 7.4$  GeV<sup>2</sup>. `moment_B` applies the same constraint without the  $F_1$  factor and yields amplitudes a factor of 2–8 lower; the A-versus-B spread is the dominant ansatz systematic, and both curves appear on every plot.

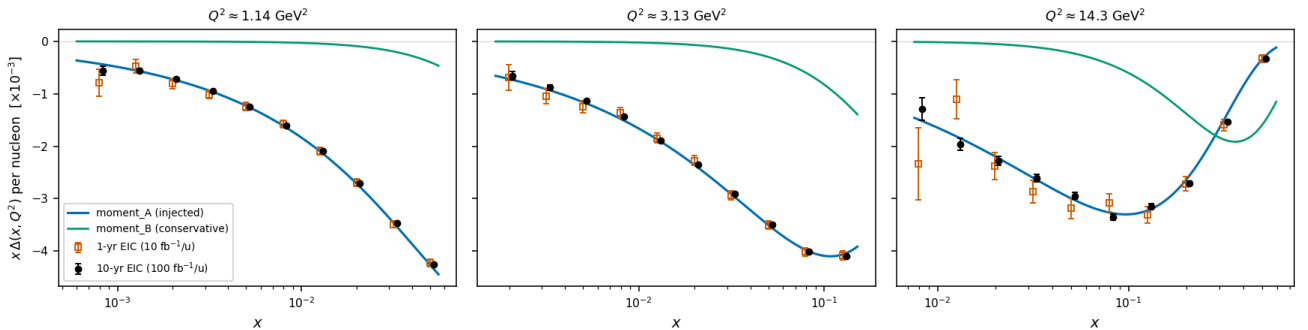
**Status of the normalization.** Transferring a single-hadron bag moment to a per-nucleon nuclear structure function has no precedent in the literature; both the normalization and the  $x$  dependence are labeled scenario choices, bracketed by `moment_B` from below and by the possible nuclear enhancement from above. The projections demonstrate that the measurement has the precision to decide: under the moment constraint, the  $x$  dependence separates interpretations A and B bin by bin.

Nuclear gluonometry, transversely tensor-polarized  ${}^6\text{Li}$ ,  $e(10) \times 6\text{Li}(50/u)$ ,  $P_{zz} = 0.60$   
moment\_A(variant=mid\_x, c=-0.012, A=-0.2941, q2\_ref=3.913, dilution=0.3333);  $\phi'$  at 1 yr (10  $\text{fb}^{-1}/u$ ), amplitudes 1 & 10 yr — stat. only; bkg & tensor RC unquantified (plans/0)



**Money plot 5 — inclusive gluonometry** ( $e10 \times {}^6\text{Li}50$ ,  $P_{zz} = 0.60$ ,  $\Delta$ : moment\_A, dilution 1/3). Left 2x2:  $\phi'$  pseudo-data with 1-year statistical error bars in the four sweet-spot super-bins ( $Q^2 = 1.1 \rightarrow 14 \text{ GeV}^2$ ,  $x \approx 0.02\text{--}0.14$ ); injected curve (blue), binned fit (vermillion); annotations give  $\delta A$  for both programs. Right: extracted amplitude vs  $x$  with 1-year (open) and 10-year (filled) error bars against the moment\_A, moment\_B, and toy- $10^{-3}$  curves. Amplitudes are  $(0.7\text{--}1.2) \times 10^{-2}$  at the sweet spots against  $\delta A = (1.5\text{--}4.5) \times 10^{-4}$  in year 1, i.e. every bin  $\geq 21\sigma$ , so the A-versus-B spread is resolved by the  $x$  dependence. Only the scattered electron is detected.

Extracted double-helicity-flip  $\Delta(x, Q^2)$  of  ${}^6\text{Li}$ ,  $e(10) \times 6\text{Li}(50/u)$ ,  $P_{zz} = 0.60$   
 $\hat{\Delta} = \hat{A} y^2 D_{\phi}/(1-y)$  per bin (model bin-centering; dilution 1/3 incl.; stat. only, bkg & tensor RC unquantified); area = S-S moment  $-0.012 \alpha_S/3$



**Money plot 7 — the extracted  $\Delta(x, Q^2)$ .** The amplitudes of money plot 5 unfolded to the structure function (model bin-centering,  $\delta \Delta = \delta A \cdot |\partial \Delta / \partial A|$ ), shown as  $x \Delta$  per nucleon at the three sweet-spot  $Q^2$  slices with independent 1-year (open) and 10-year (filled) pseudo-measurements against the moment\_A and moment\_B curves; the area under each curve is the ported bag moment ( $\div 3$  dilution). Best bins reach  $\delta \Delta \approx (0.7\text{--}1.4) \times 10^{-3}$  in year 1 on  $\Delta \approx -0.03\text{--}0.09$ .

## 4 • The coherent channel

### 4.1 • Detecting the intact ${}^6\text{Li}$ at ePIC

In coherent diffraction,  $e {}^6\text{Li} \rightarrow e' X {}^6\text{Li}(\text{g.s.})$ , the recoil retains the beam's magnetic rigidity ( $A/Z = 2$ ; the diffractive momentum loss  $x_p < 0.01$  changes it by less than 1%) and emerges at  $\theta = p_T/p \lesssim 1.7 \text{ mrad}$ . Charged particles below 5 mrad enter the far-forward system: the Bo spectrometer (5.5–20 mrad, charged tracks and photons above  $\sim 50 \text{ MeV}$ ), the Off-Momentum Detectors ( $\sim 30\text{--}60\%$  of beam rigidity), the Roman Pots at  $z \approx 26/28 \text{ m}$  ( $\theta < 5 \text{ mrad}$ ,  $\sim 60\text{--}95\%$  rigidity for off-rigidity tracks, AC-LGAD sensors with  $\sim 140 \mu\text{m}$  spatial and  $\sim 30 \text{ ps}$  timing resolution), and the ZDC for neutrals [7,8,9]. A recoil at beam rigidity remains inside the beam

envelope and is visible only in the Roman Pots, and only above the  $10\sigma$  envelope cut; at low  $|t|$  the intact recoil cannot be tagged at IP6 [10,51]. With an exponential coherent  $t$ -distribution the tag acceptance is analytic:

$$\varepsilon_{\text{tag}} = e^{-B p_{T,\text{cut}}^2} = 13.5\% \quad (p_{T,\text{cut}} = 0.20 \text{ GeV}, B = 50 \text{ GeV}^{-2}); \quad 4 \times 10^{-5} \quad (0.45 \text{ GeV})$$

**Provenance of the cuts.** The 0.20 GeV floor is the documented EIC requirement for high-acceptance (low-divergence) optics at top rigidity, corresponding to a beam divergence  $\leq 73 \mu\text{rad}$  for 275 GeV protons [7,48], and matches the one documented light-ion number: coherent  $^3\text{He}$  tagging limited to  $p_T > 0.20 \text{ GeV}$  at IR-6 [10]. The 0.45 GeV value is our derivation (150  $\mu\text{rad}$  high-divergence proton optics applied to the 300 GeV  $^6\text{Li}$  momentum), not a documented number. No beam-optics parameters have been published for any lithium beam; if the  $^6\text{Li}$  beam ran with proton-class divergence at this rigidity, the  $10\sigma$  cut would fall at 0.5–0.7 GeV and eliminate the IP6 channel. This cut is the most consequential machine input to the measurement (assumption #1, §6).

**Coherence selection at IP6 and at IR-8.** Standard EIC practice at IP6 is to establish coherence by vetoing breakup rather than by tagging the recoil: dissociation protons (rigidity 0.50) reach the Off-Momentum Detectors, neutrons the ZDC, de-excitation photons the Bo calorimeter and ZDC, and  $^3\text{He}$ -like fragments (0.75) the Roman-Pot window [51]. Direct low- $|t|$  tagging is a central argument for the second interaction region, whose lattice adds a secondary beam focus  $\sim 45 \text{ m}$  downstream where the beam size is  $\sigma \approx 0.15 \times 0.1 \text{ mm}$ , allowing Roman Pots at  $z = 44.0/45.5 \text{ m}$  to reach  $p_T \approx 0$ . The published intact-recoil efficiencies there are 47% (d), 32% ( $^3\text{He}$ ), 29% ( $^4\text{He}$ ), and 17.8% ( $^7\text{Li}$ ) at top energy, rising steeply at lower energies ( $^3\text{He}$  reaches 99.8% at  $5 \times 41$ ) [10]; no  $^6\text{Li}$  case has been published, and our interpolation is  $\approx 20\%$ . One degeneracy remains even there: a Roman-Pot hit measures only  $A/Z$ , so intact  $^6\text{Li}$ ,  $\alpha$ , and d at the same velocity are identical in trajectory and timing. The second-detector design addresses this with a proposed  $Z^2$ -scaling Cherenkov counter behind the pots, currently at the R&D stage [8]. For the IP6 AC-LGADs no charge-discrimination study exists (signal amplitude is used only for position interpolation), and the minimum-angle ion acceptance there is not yet fully established [49]. This question, open item #19, determines whether IP6 coherent-tag purity is established event by event or statistically (§5).

## 4.2 • Rate model

$$\sigma_{\text{coh}} = f_{\text{coh}}(x) \sigma_{\text{DIS}}, \quad f_{\text{coh}}(x) = \frac{f_0}{1 + (x/x_{\text{coh}})^2}, \quad f_0 = 0.04 \quad [0.02\text{--}0.08]$$

No diffractive prediction exists for a light nucleus (the lightest published case is Ca), so the  $f_0$  band brackets HERA ep diffraction (10–15% of DIS [44]) and heavy-nucleus saturation estimates (20–25% [45]). The scale  $x_{\text{coh}} = 0.01$  marks where the coherence length  $l_c = 0.105 \text{ fm}/x_p$  falls below the  $^6\text{Li}$  size. The slope  $B = 50 \text{ GeV}^{-2}$  [40–60] follows from  $B = \langle r^2 \rangle / 3$  with the matter radius (the charge radius gives  $\approx 57$  [47]; gluonic distributions may be wider [42]); the exponential form is valid below the form-factor minimum at  $|t| \approx 0.31 \text{ GeV}^2$  [11,12]. Parameterizing  $f_{\text{coh}}$  in Bjorken  $x$  rather than  $x_p$  is optimistic for high-mass diffraction, which the  $f_0$  band is intended to cover. The resulting yields are  $1.1 \times 10^7$  tagged events in year 1 and  $1.1 \times 10^8$  over ten years, with best-super-bin  $\delta A = 1.9 \times 10^{-3}$  and  $6 \times 10^{-4}$  respectively.

## 4.3 • The anchored amplitude

The  $\cos 2\phi'$  modulation of the tagged yield carries two mechanisms. The *deformation* term is scaled from the polarized-deuteron CGC calculation of Mäntysaari, Salazar, Schenke, Shen, and Zhao [13], the only published calculation of polarization-dependent azimuthal coefficients in coherent diffraction on any polarized nucleus (all forward citations checked; the group's subsequent work is unpolarized [14,15]). Their coefficients, digitized in `coherent.MANTYSAARI_A2_DEUTERON`, show  $a_2$  growing linearly in  $|t|$  with  $a_2(m=0) = -2a_2(m=\pm 1)$ , arising from the 6% D-wave modulating the diffractive slope. For a fill with populations  $p_m$  the ensemble coefficient is proportional to  $P_{zz}$ :

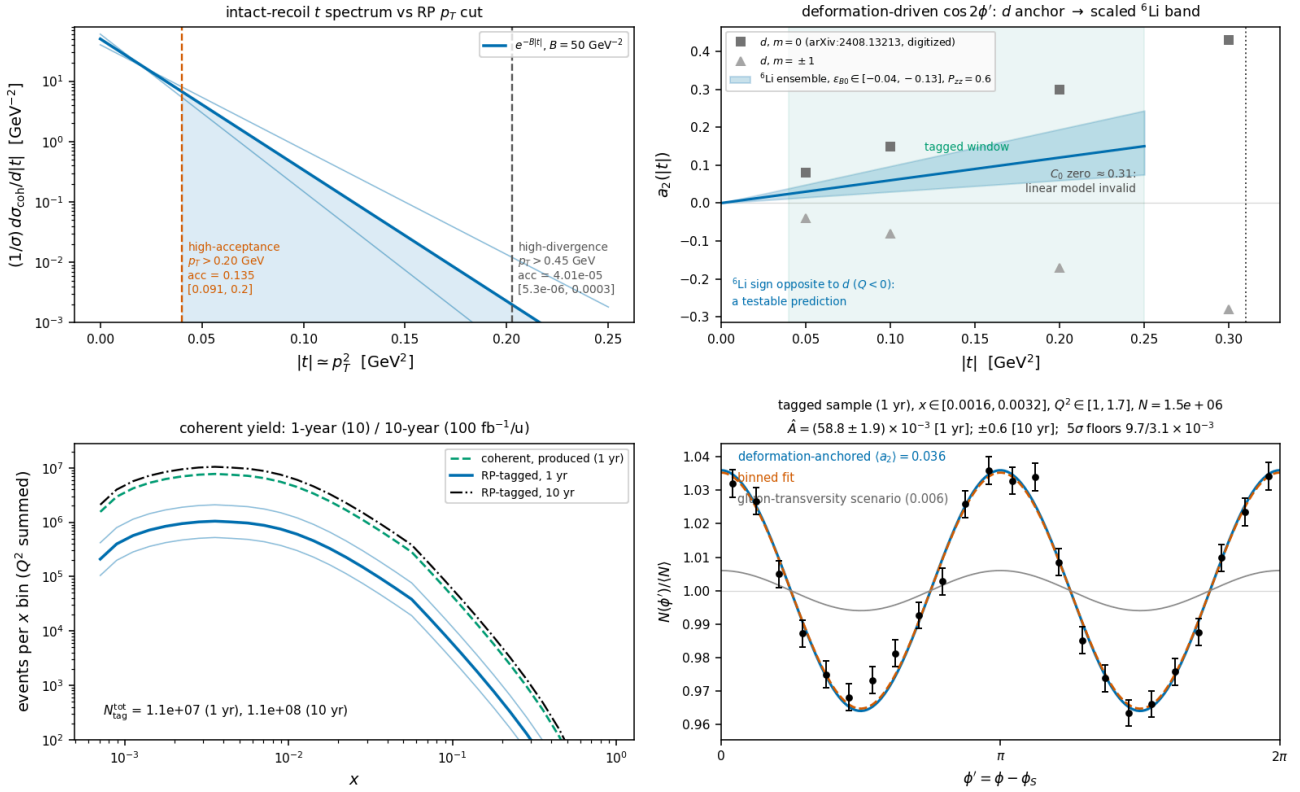
$$a_2(t; P_{zz}) = -\frac{P_{zz}}{4} \varepsilon_{B0} B |t|, \quad \langle a_2 \rangle_{\text{tag}} = a_2(p_{T,\text{cut}}^2 + 1/B) = 0.036 \quad [0.018\text{--}0.059]$$

**Convention note (added 2026-08-24).** The anchor's Eq. (9) expands  $d^2\sigma/d\Phi d|t| \propto 1 + 2\Sigma_n a_n e^{in\Phi}$ , with  $\Phi$  the azimuth of the produced vector meson — the momentum transfer, i.e. the recoil — relative to the polarization axis in the  $\gamma^*d$  frame. The  $\cos 2\Phi$  modulation coefficient is therefore  $2a_2$ , and the  $\langle a_2 \rangle$  quoted here and injected in money plot 6(d) understates the deformation modulation by a factor two (0.072 rather than 0.036 at  $P_{zz} = 0.6$ ); this report's coherent numbers are conservative by  $\times 2$  and are kept in the  $a_2$  convention until the WP5 curves are redone. The modulation is one of the *recoil* azimuth  $\varphi_t - \varphi_S$ , not of the electron azimuth used as the axis of panel (d); the reconstructed-level companion (reconstruction-chain report §7, money plot 6R) injects the corrected coefficient in the recoil azimuth and fits both azimuths.

$\varepsilon_{B0} \equiv \Delta B_0/B$  is scaled from the deuteron value of 0.21 using two indicators that disagree: the relative quadrupole deformation  $Q/\langle r^2 \rangle$  of  ${}^6\text{Li}$  is about one fifth of the deuteron's and opposite in sign, while the  $\alpha$ -d asymptotic D/S ratio is disputed (a transfer measurement gives  $\eta = +0.0003(9)$ , consistent with zero [16], whereas older sub-Coulomb analyses favored  $\eta \approx -0.01$ ). We adopt  $\varepsilon_{B0} \in -(0.04-0.13)$  with default  $-0.08$ , leaving near-zero deformation open; the  $5\sigma$  floors cover its detection or exclusion. The width of the band reflects genuine theoretical uncertainty: even exact few-body calculations miss the  ${}^6\text{Li}$  quadrupole cancellation by a factor of  $\approx 3$  [17], and the gluonic deformation need not share the cancellation observed in the charge sector. The sign flip relative to the deuteron is a testable prediction ( $Q({}^6\text{Li}) < 0$ ). Tensor modulations of comparable size are established at these  $|t|$ : e-d  $T_{20} = -(0.1-0.3)$  [19] and HERMES  $A_{zz}(d) \approx -1\%$  at  $x \approx 0.01$  [20], the latter attributed to coherent double scattering [21–23]. The flat gluon-transversity term is separately bounded at  $3 \times 10^{-3} - 10^{-2}$ , from the bare lattice  $\phi$  transversity [25] divided by the  $\sim 10\times$  NPLQCD deuteron suppression [27] (both at unphysical pion masses), with the maximal Drell–Yan scenario [5] as the ceiling. The equal-rate population average behind  $\langle a_2 \rangle$  carries a one-sided  $\times 0.73$  rate-weighting systematic (audited; within the  $\varepsilon_{B0}$  band).

**Separation of the two mechanisms.** The mechanisms differ in shape: deformation is linear in  $t$ , sign-flipped relative to the deuteron, and expected to weaken toward small  $x_P$  (JIMWLK evolution washes out deformation [15]); exotic glue is flat in  $t$  and persists at small  $x$ , where it connects to the elliptic-gluon program [6]. In a two-component fit  $a_2(t) = c_{\text{def}}|t| + a_g$  per  $x_P$  bin, both components exceed the statistical floor by  $\geq 5\sigma$  at  $100 \text{ fb}^{-1}/u$ . Incoherent breakup is  $m$ -state-blind in this mechanism [13]: it dilutes the modulation but cannot mimic it.

Coherent  $e^6\text{Li} \rightarrow e'X^6\text{Li(g.s.)}$ , transversely tensor-polarized,  $e(10) \times 6\text{Li}(50/u)$ ,  $P_{zz} = 0.60$   
 nario rates; modulation anchored on the polarized- $d$  CGC calculation, PLB 858:139053 -- plans/06 SS6.4b; statistical only, backgrounds/purity and tensor RC unquanti



**Money plot 6 — coherent channel,  $e^6\text{Li} \rightarrow e'X^6\text{Li(g.s.)}$**  (same beam,  $P_{zz} = 0.60$ ). Top left: recoil  $t$ -spectrum vs both optics'  $p_T$  cuts with the slope band. Top right: digitized deuteron  $a_2$  per  $m$ -state [13] vs the scaled  $^6\text{Li}$  ensemble band (sign flip predicted); tagged window and form-factor-zero exclusion marked. Bottom left: tagged yield vs  $x$  for both programs with the  $f_0$  band ( $N_{\text{tag}} \approx 1.1 \times 10^7 / 1.1 \times 10^8$ ). Bottom right:  $\phi'$  pseudo-data (1-year statistics) at the anchored  $\langle a_2 \rangle = 0.036$  with the flat gluon-transversity scenario (gray); extraction  $\hat{A} \pm 1.9 \times 10^{-3}$  (1 yr),  $\pm 0.6 \times 10^{-3}$  (10 yr).

## 5 • Backgrounds for the coherent tag (full budget: plans/06)

| BACKGROUND                                                                                                                                                 | WHY IT FAKES THE TAG                                                                                                                                           | HANDLE                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><math>\alpha + d</math> breakup</b> (threshold 1.474 MeV; the 2.186 MeV $3^+$ state, $\Gamma = 24 \text{ keV}$ , decays $\sim 100\%$ $\alpha + d$ [39]) | Both fragments at $R = 1.000$ , same trajectory and velocity as intact $^6\text{Li}$ : tracking and ToF blind; only $dE/dx$ ( $\propto Z^2: 9/4/1$ ) separates | <b>RP charge discrimination — open question #19</b> : no study exists for the IP6 AC-LGADs (amplitude is used for position; the documented Z-ID is the proposed IR-8 Cherenkov [8]). Fallback: two-component $ t $ fit (coherent $e^{-50 t }$ vs $\sim 10\times$ flatter incoherent; crossover at $0.05\text{--}0.2 \text{ GeV}^2$ inside the window [41,42]); $e + \text{Pb}$ benchmark: 80–99% incoherent rejection [43] |
| Nucleon-emission breakup: $\alpha + p + n$ (3.70 MeV), $^3\text{He} + t$ (15.79 MeV) [39]                                                                  | A fragment could pass unnoticed                                                                                                                                | $p \rightarrow \text{OMD}$ ( $R = 0.50$ ), $n \rightarrow \text{ZDC}$ , $^3\text{He} \rightarrow \text{RP window}$ ( $R = 0.75$ ) [51]: vetoable; inefficiency is second-order (BeAGLE study to quantify)                                                                                                                                                                                                                  |
| $^6\text{Li}^*$ 3.5629 MeV ( $0^+$ , $T = 1$ , $\Gamma = 8.2 \text{ eV}$ ) [39]                                                                            | Particle-stable ( $\alpha + d$ parity-forbidden, $\Gamma_{\alpha} \leq 6 \times 10^{-7} \text{ eV}$ ); $\gamma$ -decays to a real $A = 6$ , $Z = 3$ track      | Isoscalar exchange cannot drive $T = 0 \rightarrow 1$ ; the boosted $0.1\text{--}0.4 \text{ GeV}$ $\gamma$ lands mostly in the B0 calorimeter (photons $\geq 50 \text{ MeV}$ , $5.5\text{--}20 \text{ mrad}$ [9]), tail in the ZDC. The 5.366 MeV $2^+ T = 1$ state always emits a nucleon. Every $T = 1$ state is $\gamma$ - or nucleon-vetoable; only $T = 0$ states feed the blind $\alpha + d$ channel                 |
| Bethe–Heitler / QED Compton                                                                                                                                | Intact nucleus by construction                                                                                                                                 | Calculable ( $t$ -dependence = charge $\text{FF}^2$ [11,12]); subtract and reuse as an RP-acceptance candle; the charge-quadrupole $\cos 2\varphi$ is $< 10^{-3}$ for $ t  < 0.15 \text{ GeV}^2$ [12,17]                                                                                                                                                                                                                   |
| Beam-gas / halo; $e'$ -side fakes                                                                                                                          | Accidental $R \approx 1$ tracks; $\pi$ / pair-symmetric $e'$ at high $y$                                                                                       | Non-colliding bunches, $\sim 30 \text{ ps}$ timing, vertex association; standard DIS $e'$ ID plus ( $e'$ , recoil) kinematic consistency                                                                                                                                                                                                                                                                                   |

Concerning extraction systematics: the modulation is self-normalized within a single fill, so relative-luminosity systematics cancel;  $\phi$ -acceptance holes are handled by the binned fit; the polarimetry scale  $\delta P_{zz}/P_{zz} \approx 3\%$  is common with  $A_{zz}$ ; and radiative corrections for tensor observables have not been calculated, which is noted on every plot. At the level of interpretation, a  $\Delta\Delta$ -isobar admixture can mimic  $\Delta$  at leading twist [34] and conventional double scattering generates the low- $x$   $b_1$  [21–23]; the  $Q^2$  lever arm and the  ${}^6\text{Li}/d$  comparison discriminate against both.

**The  ${}^6\text{Li}$  null test.** Deformation-driven coherent modulations, the polarized-deuteron imaging signal [13], are bounded on  ${}^6\text{Li}$  by its anomalously small quadrupole moment,  $Q = -0.0806(6) \text{ fm}^2 \approx Q({}^7\text{Li})/50$  [35]. A sizable flat- $\text{in-}t \cos 2\phi$  on  ${}^6\text{Li}$  therefore cannot be a shape effect; the mechanisms that are entangled for the deuteron separate in the  ${}^6\text{Li}/d$  comparison.

## 6 • Critical assumptions

| #  | ASSUMPTION                                                                                                                                                                                               | IMPACT IF WRONG                                                                                                      | REPLACEMENT PATH                                                                                      |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| 1  | RP $p_T$ cuts: 0.20 GeV is the documented top-rigidity envelope scale ( ${}^3\text{He}$ precedent [10]); 0.45 GeV is our derivation; no Li optics exist — proton-class divergence would give 0.5–0.7 GeV | Tag acceptance anywhere from 13.5% to negligible at IP6; the physics case shifts to IR-8                             | Li beam-optics parameters from EIC C-AD; the IR-8 secondary focus removes the cut entirely [10]       |
| 2  | Scenario coherent fraction $f_0 = 0.04$ [ $\times 2/\div 2$ ], parameterized in Bjorken $x$                                                                                                              | Coherent yields scale linearly; $\delta A$ as $1/\sqrt{f_0}$                                                         | Real diffractive model — open question #18; concrete ask: an IP-Glasma $\alpha$ -d run [13]           |
| 3  | Bag-moment normalization of $\Delta$ ported to per-nucleon ${}^6\text{Li}$ with 1/3 dilution; $x$ -shape ours                                                                                            | Overall $\Delta$ scale; literature brackets both directions (NPLQCD suppression [27] vs binding enhancement [50])    | Lattice moment for a light nucleus; theory engagement (plans/04)                                      |
| 4  | $\epsilon_{B0}$ scaling from the deuteron (two dials disagreeing by $\times 3$ )                                                                                                                         | Deformation amplitude anywhere in the band; sign flip robust                                                         | The same IP-Glasma $\alpha$ -d calculation; band quoted on every plot                                 |
| 5  | Event-by-event Z-ID in the RPs is NOT assumed (purity via $ t $ -shape fit)                                                                                                                              | If Z-ID exists purity improves; if the incoherent slope model is off, the fit systematic grows                       | #19 answer from the ePIC far-forward group; BeAGLE incoherent spectra                                 |
| 6  | Linear $a_2(t)$ valid only for $ t  < 0.2 \text{ GeV}^2$ ; form-factor zero at $0.31 \text{ GeV}^2$ excluded, not modeled                                                                                | Upper tagged window carries unmodeled structure                                                                      | Form-factor-level model of the dip region (deuteron analog in [13])                                   |
| 7  | TOY structure functions; no target-mass terms; $g_2 = g_2^{WW}$                                                                                                                                          | Rates to $\times 1.5$ ; shapes are scenarios                                                                         | PDF-grid backends exist; digitized theory curves scheduled (plans/02)                                 |
| 8  | All luminosity in the transverse-tensor fill; $P_{zz} = 0.6$ in-ring placeholder                                                                                                                         | $N$ scales with the run-plan share; amplitude with $P_{zz}$                                                          | Run-plan bookkeeper supports splits; source demonstrations reach $\sim 95\%$ of maximal $P_{zz}$ [37] |
| 9  | Impulse approximation; no FSI; tensor radiative corrections unknown                                                                                                                                      | FSI distorts the recoil spectrum at larger $ t $ ; RC unquantified                                                   | Theory engagement (plans/04); the QED tail is calculable and doubles as calibration                   |
| 10 | Equal-rate population average for $\langle a_2 \rangle$ ; $t_{\text{min}}$ neglected in the tag acceptance                                                                                               | One-sided $\sim 27\%$ (rate weighting) and $\sim 10\%$ ( $t_{\text{min}}$ ) — inside the $\epsilon_{B0} / f_0$ bands | Rate-weighted ensemble model (the IP-Glasma run supersedes both)                                      |

## 7 • Next steps

**Near term:** BeAGLE incoherent  ${}^6\text{Li}$  spectra to quantify the veto efficiency and the  $|t|$ -fit systematics (plans/02); VMC  $\alpha$ -d/ $\alpha$ -t overlaps to replace the two-parameter radial forms (plans/04); pseudo-experiment closure with a two-dimensional ( $\phi$ ,  $\phi_t$ ) acceptance map (plans/03); and HepMC3 output with an ePIC chain test (plans/05), so that these projections can be repeated at reconstructed level.

**Theory:** a rerun of the IP-Glasma polarized-deuteron setup [13] with an  $\alpha$ -d cluster density, the highest-leverage input, which would replace both  $\epsilon_{B0}$  indicators and the  $f_0$  band at once; a lattice  $\Delta$  moment for a light

nucleus; the spin-3/2 structure-function basis for  ${}^7\text{Li}$ ; tensor-observable radiative corrections; and the sum-rule connection between transversity-GPD moments and measured tensor form factors [30].

**Machine and detector:** lithium beam-optics parameters, which set the IP6 p\_T cut (assumption #1); an answer to open item #19, whether the RP AC-LGAD amplitude can separate  $Z = 3, 2, 1$  at identical rigidity and velocity; and the IR-8 secondary focus with its proposed  $Z^2$  Cherenkov [8], which extends the tag toward  $p_T \approx 0$  ( ${}^7\text{Li}$ : 17.8% published;  ${}^6\text{Li}$  not yet studied [10]). The coherent- ${}^6\text{Li}$  program adds one more physics case for the second interaction region. Off-momentum halo determines how closely the pots approach the beam, and therefore the cut itself.

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checks of the Sather–Schmidt moment usage and the far-forward detection chain. Reference numbering is shared with earlier revisions; retired entries are omitted. Target venue: INT program on polarized ion beams at the EIC, March 22 – April 2, 2027.